

MUHAMMAD OWAIS

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RESEARCH INTERESTS

PhD Researcher, Deep Learning for Vision and Speech; Multimodal Representation Learning; Transformer-Based Models; Vision Transformers; CNN-based Models; Robustness and Generalization in Neural Systems.

EDUCATION

• PhD. Computer Science (In Progress)

Chengdu, China

University of Electronic Science and Technology (UESTC)

• MS / MPhil in Computer Science

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)

Swabi, Pakistan

Jan 2023 – June 2025

◦ CGPA: 3.17 / 4.00

◦ Thesis: *Deepfake Audio Detection in Low-Resource Languages: A Case Study of Urdu*

• BS in Computer Science

Abdul Wali Khan University Mardan

Mardan, Pakistan

Sep 2018 – Aug 2022

◦ CGPA: 3.27 / 4.00

PUBLICATIONS

[J1] Owais, M., Jadoon, K. K., Sandhu, A. I., Ali, Z., Mahmood, Z., Yahya, M., & Wahid, A. **Deepfake Audio Detection in Low-Resource Languages: A Case Study of Urdu**. *IEEE Access*, 2026. DOI:

<https://doi.org/10.1109/ACCESS.2026.3654621>

[J2] Owais, M., Sarwar, K., Jadoon, K. K., & Yousaf, J. **Emotion Detection in Urdu Speech: A Deep Hybrid Learning Approach**. *International Journal of Speech Technology (Q1)*, 28(3), 761–772, 2025. DOI:

<https://doi.org/10.1007/s10772-025-10212-1>

[J3] Co-Authored – **Multimodal Fusion of Vision Transformers and Clinical Language Models for Medical Visual Question Answering**.

Submitted to *Journal of Medical Systems*

EXPERIENCE

• City University of Science and Information Technology

Lecturer, Computer Science

Sep 2025 – June 2026

Peshawar, Pakistan

◦ Teaching undergraduate courses in Programming, Artificial Intelligence, and Machine Learning.

◦ Supervising final-year research projects in applied deep learning.

• Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

Graduate Teaching & Research Assistant

Jan 2023 – Jan 2025

Swabi, Pakistan

◦ Assisted courses: AI, Machine Learning, Python, NLP, OOP, Databases (theory & labs).

◦ Conducted research in Deep Learning, Speech Processing, NLP, and Computer Vision.

◦ Contributed to experimental design, ablation studies, and journal publications.

RESEARCH PROJECTS

• Breast Ultrasound Tumor Classification (Benign vs. Malignant)

2025

CNN, Transfer Learning, Breast Ultrasound Dataset

◦ Developed a CNN-based deep learning model to classify benign and malignant tumors in breast ultrasound images.

◦ Applied transfer learning and data augmentation techniques to enhance robustness, generalization, and classification performance.

• Handwritten Character Recognition using CNN and Vision Transformer

2025

CNN, Vision Transformer, EMNIST

◦ Designed a hybrid CNN–Vision Transformer architecture for handwritten character recognition.

◦ Leveraged spatial feature extraction and attention mechanisms to improve recognition accuracy.

• Grape Leaf Disease Detection using Deep Learning

2025

CNN, Transfer Learning, Data Augmentation

- Developed a CNN-based model to classify grape leaf diseases with high accuracy.
 - Applied data augmentation and transfer learning to enhance model robustness and generalization.
- **Student Data Preprocessing and Profiling for University Campus Records** 2024
Data Preprocessing, Data Profiling, Python
 - Designed a preprocessing pipeline for cleaning and profiling large-scale student records.
 - Addressed missing values, outliers, and inconsistencies to improve analytical reliability.
 - **Comparative Analysis of Deep Learning Architectures for Fake News Detection** 2023
CNN, RNN, LSTM, Attention
 - Investigated multiple deep learning architectures for fake news classification.
 - Improved detection performance through model fine-tuning and feature analysis.
 - **Comparative Analysis of GAN and DCGAN for Multilingual Alphabets and Numeric Digits Generation** 2023
GAN, DCGAN, Image Generation
 - Conducted a comparative study of GAN and DCGAN for multilingual alphabet and digit generation.
 - Evaluated image quality, diversity, and computational efficiency across models.

CONFERENCES & WORKSHOPS ATTENDED

- **2nd International Symposium on AI & Quantum Computing**, GIKI, Pakistan Apr 2024
- **1st National Workshop on AI in Healthcare & Drug Discovery (AI-h2D)**, Pakistan May 2024
- **Young Scientist Workshop on Applied AI**, GIKI, Pakistan Sep 2024

HONORS AND AWARDS

- **Graduate Assistantship Scholarship** 2023 – 2025
GIKI, Pakistan
 - Merit-based scholarship covering tuition and stipend for teaching and research duties.

SKILLS

- **Programming & Scientific Computing:** Python, C++, SQL
- **Deep Learning & Representation Learning:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, CNNs, Transformers, Vision Transformers (ViTs), LSTMs
- **Speech & Natural Language Processing:** MFCC, LFCC, Mel-Spectrograms, Automatic Speech Recognition (ASR) Models
- **Computer Vision:** Image Classification, Feature Extraction, Data Augmentation, Transfer Learning
- **Research Skills:** Experimental Design, Literature Review, Ablation Studies, Model Evaluation, Error Analysis, Reproducibility, Academic Writing
- **Tools & Platforms:** Git, Linux, LaTeX, Overleaf

REFERENCES

1. **Dr. Ali Imran Sandhu**
Assistant Professor, GIKI
Email: aliimran.sandhu@giki.edu.pk
Relationship: MS Thesis Supervisor
2. **Dr. Khurram Khan Jadoon**
Assistant Professor, GIKI
Email: khurram.jadoon@giki.edu.pk
Relationship: MS Thesis Co-Supervisor
3. **Prof. Dr. Muhammad Arif Shah**
Prof. and Dean Faculty of Computing and Numerical Sciences, CUSIT
Email: dean.cns@cusit.edu.pk
Relationship: Academic Dean